# UNIT: 1 Assignment

① What is Microprocessor? What is the importance of microprocessor in a computer system?

⇒ A microprocessor is a multipurpose, programable, clock-driven, register-based electronic chip that ~~(functions as)~~ accepts binary data as input reads binary instructions from memory, processes data according to instructions and provide result as output. It functions as the CPU of a computer system.

The microprocessor serves as the primary brain of a microcomputer. It's importance lies in its ability to:

(i) Execute Instructions: It sequentially fetches, decodes and executes software instructions to process data.

(ii) Central Control: It's control unit generates crucial timing and control signals to communicate, co-ordinate and manage all peripherals.

(iii) Perform Complex Calculations: It features an ALU capable of carrying out high-speed arithematic & logical operations.

(iii) Deliver Efficiency: It introduces a highly reliable, cost-effective, and versatile architecture that allows the same hardware chip to perform vastly different applications simply by changing the software program

2. Explain about Evolution of microprocessor from the first stage processor to latest processor.

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② Explain about the Evolution of microprocessor from first stage to latest processor.

| Generation | Timeline | Architectural Feature | Technology | Example |
|---|---|---|---|---|
| First | 1971-1973 | 4-bit and some 8 bit | PMOS technology, slower speeds | Intel 4004, Intel 8008 |
| Second | After 1973 | 8-bit or 16-bit | NMOS, offered faster speed, higher density and expanded instruction set | Intel 8080, Motorola M680 |
| Third | After 1978 | 16-bit | HMOS, strong processing capabilities and massive instruction sets | Intel 8086 |
| Fourth | Introduced 1980 | Generally 32-bit | CMOS, high operating speeds, complex functional capabilities etc | Intel 80386, MC-88100 |
| Fifth | Modern Era | ~~Designs suo~~ Decoupled Super Scalar Processing | Designs surpassed 10 million transistor, low margin high volume chips | Intel i5, i7 etc |

(3) What are the basic units of microprocessor? Explain about basic organization of microprocessor and it's components along with diagram

=> A microprocessor operates using an internal architecture split into three core units, communicating with external devices via system bus.

* Basic Units of a microprocessor

(i) Arithematic Logic Unit (ALU): Performs arithematic calculations and logical operations. The results are stored either in register or the memory.

(ii) Register Array: A collection of temporary storage registers that hold binary data and memory address during program execution.

(iii) Control Unit: It is the coordinator of the processor. It interprets and generates the exact timing and control signals required to manage data flow between microprocessor, memory and peripherals.

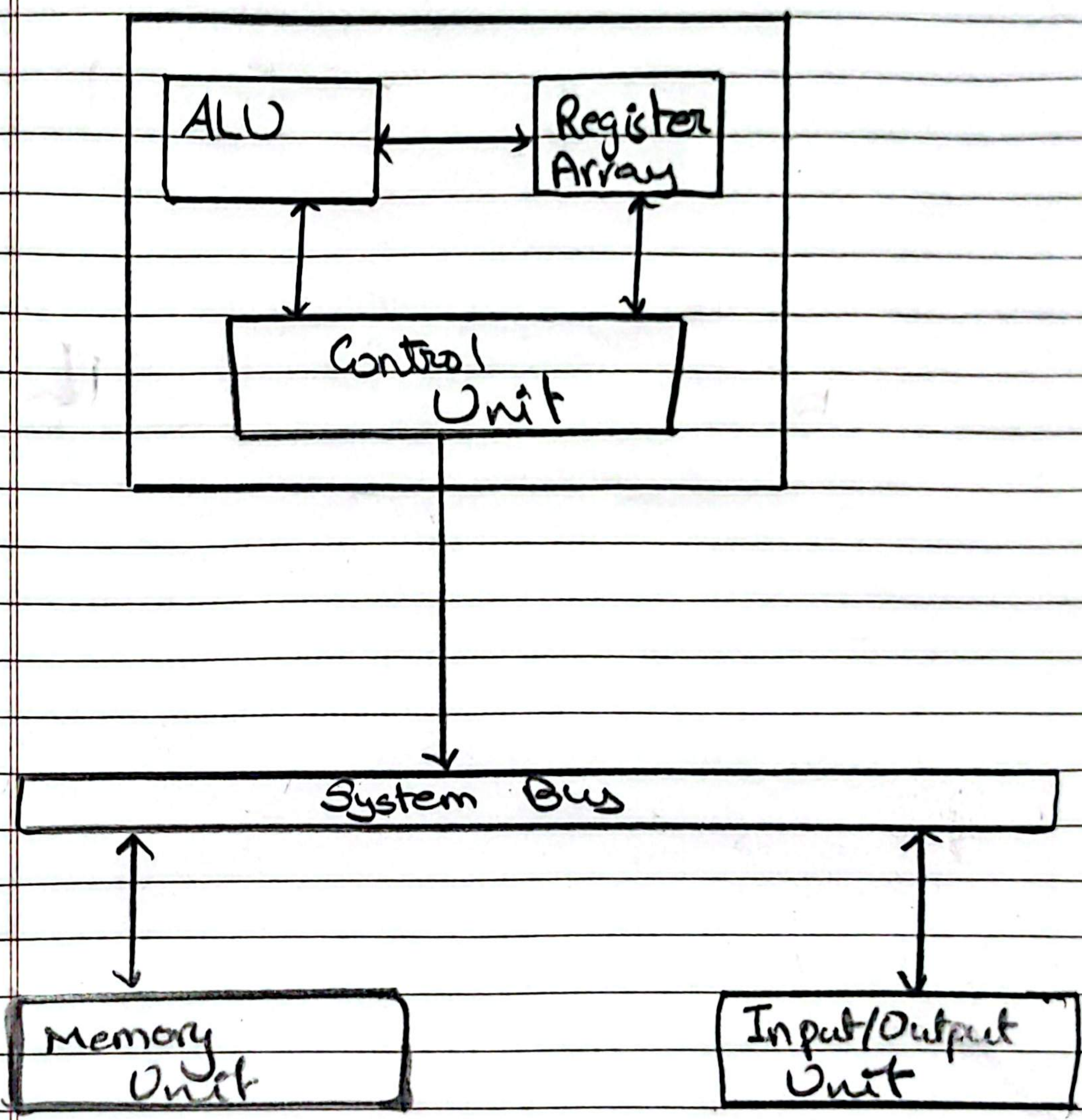


Fig: Diagramatic Representation of Basic Organization of Microprocessor